EXPERIMENT 7 – K-MEANS CUSTOMER CLUSTERING

AIM

To segment mall customers by Annual Income and Spending Score using K-Means clustering.

ALGORITHM

1. Load and inspect Mall_Customers.csv.
2. Select Annual Income and Spending Score and standardise them.
3. Use the Elbow Method for K=1 to 10.
4. Train K-Means using the chosen K and assign cluster labels.
5. Plot clusters and centroids; report counts and means; save labelled data.

PROGRAM In [ ]

```
import pandas as pd, matplotlib.pyplot as plt, seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
import os

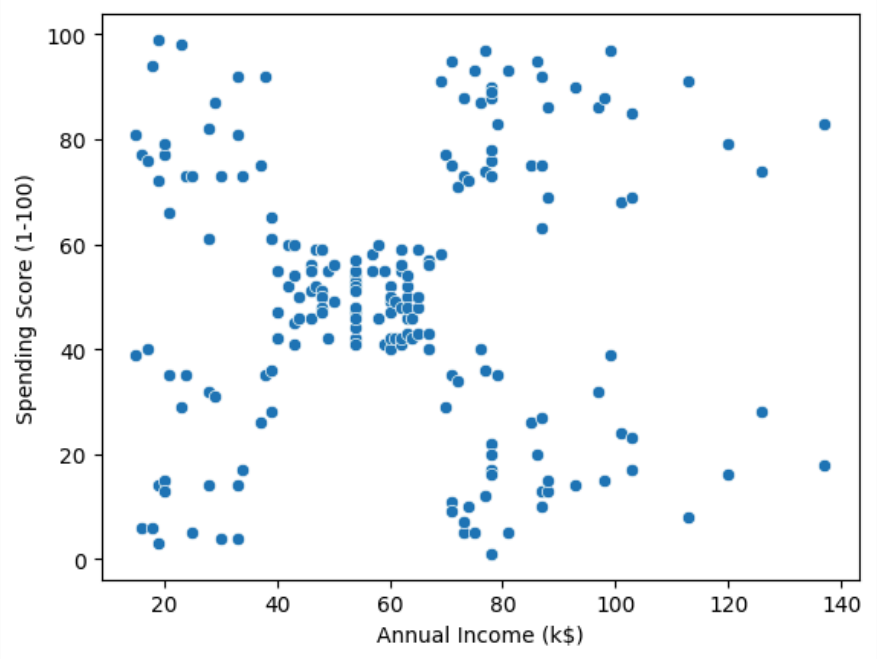
# Automatically download the dataset if it's missing using a verified fallback URL
filename = "Mall_Customers.csv"
if not os.path.exists(filename):
    print(f"{filename} not found. Downloading from GitHub...")
    url = "https://raw.githubusercontent.com/tirthajyoti/Machine-Learning-with-Python/master/Da
tasets/Mall_Customers.csv"
    try:
        df_temp = pd.read_csv(url)
        # Ensure the column names match what the script expects
        df_temp = df_temp.rename(columns={
            'Annual Income (k$)': 'Annual Income (k$)',
            'Spending Score (1-100)': 'Spending Score (1-100)'
        })
        df_temp.to_csv(filename, index=False)
        print("Successfully downloaded and saved Mall_Customers.csv")
    except Exception as e:
        print("Download failed, please upload Mall_Customers.csv manually:", e)

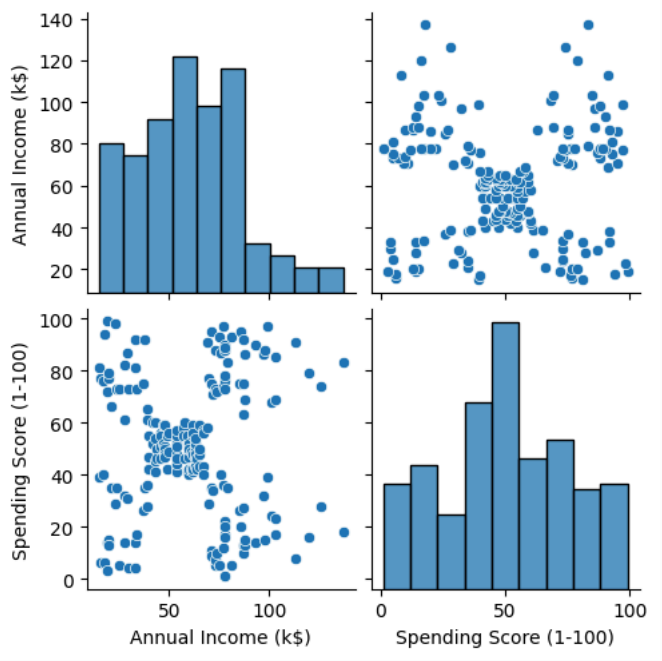
df=pd.read_csv("Mall_Customers.csv"); print(df.head(),df.tail(),df.shape,df.isnull().sum(),df.d
escribe())
cols=["Annual Income (k$)","Spending Score (1-100)"]; X=df[cols]; scaler=StandardScaler(); Z=sc
aler.fit_transform(X)
sns.scatterplot(data=df,x=cols[0],y=cols[1]); plt.show(); sns.pairplot(df[cols]); plt.show()
inertia=[]
for k in range(1,11): inertia.append(KMeans(n_clusters=k,random_state=42,n_init=10).fit(Z).iner
tia_)
plt.plot(range(1,11),inertia,marker="o"); plt.xlabel("K"); plt.ylabel("Inertia"); plt.show()
k=5 # Choose K based on the elbow plot
km=KMeans(n_clusters=k,random_state=42,n_init=10); df["Cluster"]=km.fit_predict(Z)
centres=scaler.inverse_transform(km.cluster_centers_)
sns.scatterplot(data=df,x=cols[0],y=cols[1],hue="Cluster",palette="tab10")
plt.scatter(centres[:,0],centres[:,1],marker="X",s=200,color="black"); plt.show()
print("Centroids:",centres); print(df.Cluster.value_counts().sort_index()); print(df.groupby("C
```

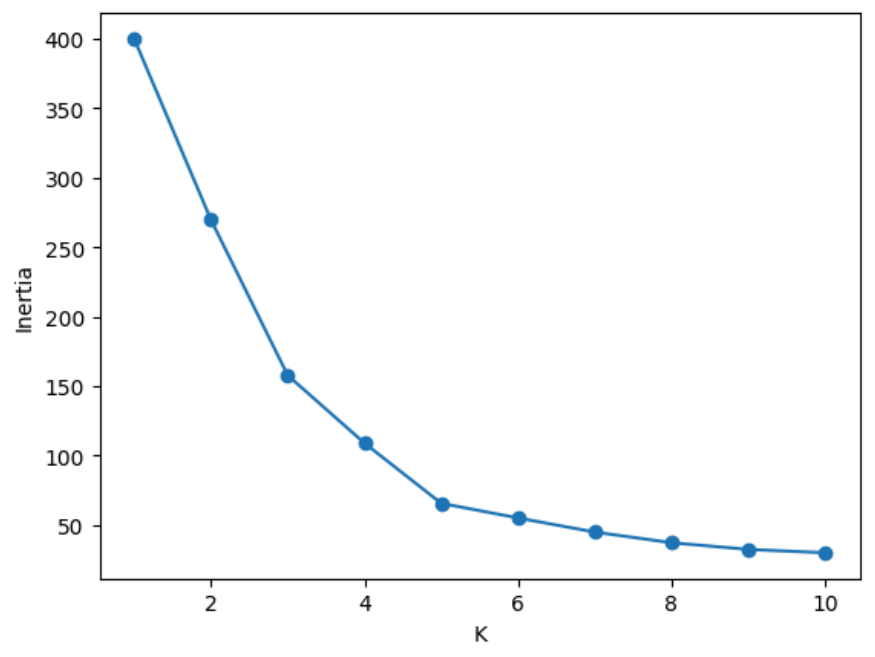
```
luster")[cols].mean())
df.to_csv("Mall_Customers_Clustered.csv",index=False)
```

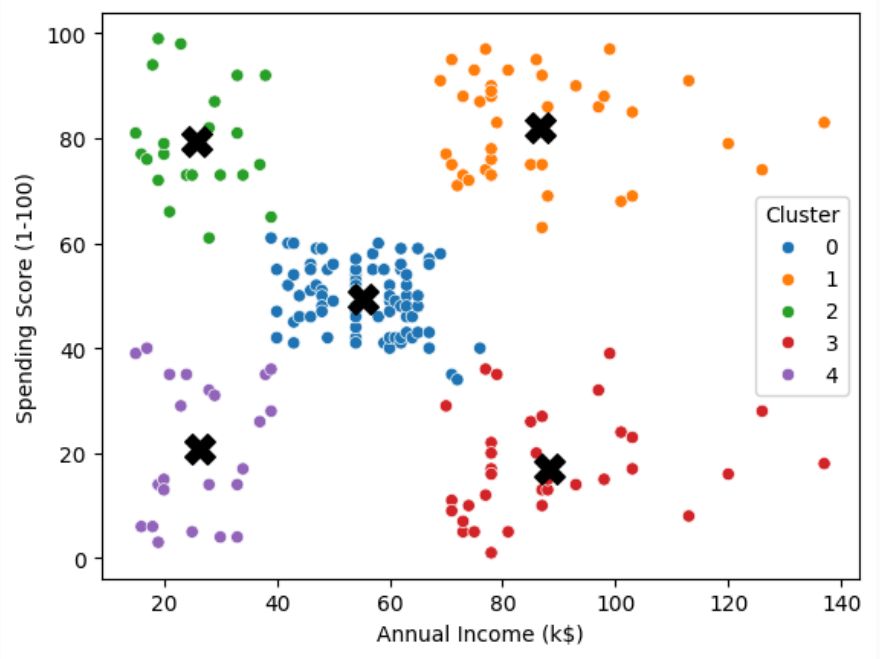
OUTPUT

```
Mall_Customers.csv not found. Downloading from GitHub...
Successfully downloaded and saved Mall_Customers.csv
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40      CustomerID  Gender
Age  Annual Income (k$)  Spending Score (1-100)
195         196  Female   35                  120                      79
196         197  Female   45                  126                      28
197         198    Male   32                  126                      74
198         199    Male   32                  137                      18
199         200    Male   30                  137                      83 (200, 5) CustomerID
0
Gender                     0
Age                        0
Annual Income (k$)         0
Spending Score (1-100)     0
dtype: int64        CustomerID         Age  Annual Income (k$)  Spending Score (1-100)
count  200.000000  200.000000          200.000000              200.000000
mean   100.500000   38.850000           60.560000               50.200000
std     57.879185   13.969007           26.264721               25.823522
min      1.000000   18.000000           15.000000                1.000000
25%     50.750000   28.750000           41.500000               34.750000
50%    100.500000   36.000000           61.500000               50.000000
75%    150.250000   49.000000           78.000000               73.000000
max    200.000000   70.000000          137.000000               99.000000
```




Annual Income (k$)
Spending Score (1-100)
Annual Income (k$)
Spending Score (1-100)


Inertia
K




```
Centroids: [[55.2962963  49.51851852]
 [86.53846154 82.12820513]
 [25.72727273 79.36363636]
 [88.2        17.11428571]
 [26.30434783 20.91304348]]
Cluster
0    81
1    39
2    22
3    35
4    23
Name: count, dtype: int64
         Annual Income (k$)  Spending Score (1-100)
Cluster
0                 55.296296               49.518519
1                 86.538462               82.128205
2                 25.727273               79.363636
3                 88.200000               17.114286
4                 26.304348               20.913043
```

RESULT

Thus, customers are segmented by income and spending behaviour and the resulting cluster profiles are summarised.